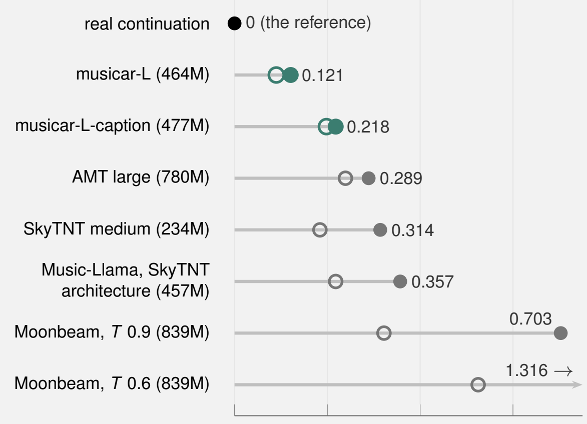
What was found and tested

The evidence behind the captioned music model (477M; the model itself is on the first sheet, *Captioned music model*): how it compares with other open MIDI models, what F1 and its caption phase changed, and the pilots that chose each part. **Teal** = what the model uses / our models; **grey** = the alternatives; pink = captions.

① How it compares with other open MIDI models (head-to-head, 2026-10-29 / 30)

Continuation: 32 prompts, 1,000 new notes each



● the first 1,000 new notes, ○ the first 200. Over 1,000 notes both of our models continue closer to the real music on these density and instrument-use statistics than every other open model run here; over the first 200 every model is closer, SkyTNT medium passes musicar-L-caption (0.184 vs 0.198) and musicar-L stays closest (0.090). D in units of the real rows' spread (with a floor); 0 = the real continuations.

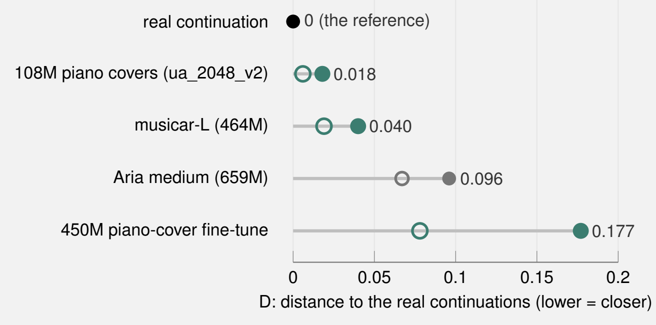
The row metrics behind D (mean over the 32 rows)

system	notes / s	instr.	out of band	top share	takeover	runaway	copies	ended	D	D, 200	params
real continuation	24.0	7.69	0.153	0.390	0	0.156	0.049	0.22	0	0	
musicar-L	23.5	6.91	0.103	0.418	0	0.125	0.144	0.31	**0.121**	**0.090**	464M
musicar-L-caption	12.8	7.78	0.170	0.415	0	0.063	0	0.53	**0.218**	0.198	477M
AMT large [15]	21.2	7.44	0.117	0.468	0.094	0.063	0.043	0.78	0.289	0.239	780M
SkyTNT medium [16]	22.9	5.34	0	0.439	0.031	0.156	0.136	0.28	0.314	0.184	234M
Music-Llama [17]	28.9	5.69	0.020	0.439	0.031	0.250	0.137	0.25	0.357	0.218	457M
Moonbeam [21], T 0.9	49.5	8.88	0.140	0.491	0.156	0.312	0.208	0.16	0.703	0.322	839M
Moonbeam, T 0.6	85.9	5.25	0.018	0.533	0.125	0.063	0.456	0	1.316	0.525	839M

Setup. Every system writes MIDI for one frozen prompt set; every file is read back from the MIDI and scored the same way, per note and per second, never per token, so the tokenizers never enter. Prompts: the 32 clean GigaMIDI validation files of the sample evaluation (128 notes, $\geq$ 3 instruments, evenly spaced). Each model runs with its authors' sampling (AMT top-p 0.98; SkyTNT and Music-Llama T 1.0, top-p 0.94, top-k 20; Moonbeam T 0.9, top-p 0.95 from its README and T 0.6, top-p 0.9 from its completion example, its 1,024-position window restarted on the last 768 notes by the harness, not the authors' scheme); ours with their presets (musicar-L T 0.9; musicar-L-caption the app's f1 preset, no caption).

Where the distance comes from. musicar-L: its top instrument plays a little more (top share 0.418 vs 0.390), it uses under one instrument fewer (6.9 vs 7.7) and it stays closer to the prompt's band (10% vs 15% out of band). musicar-L-caption: 0.14 of its 0.218 is density: it plays half as many notes per second (12.8 vs 24.0) and ends its piece inside the 1,000 notes in 53% of the rows (length training: no length asked = a short piece). AMT: one instrument takes over 9% of the rows. SkyTNT and Music-Llama: almost no new instruments (0–2% out of band vs 15% in the real music), 5.3–5.7 instruments, and Music-Llama runs away in a quarter of the rows. Moonbeam plays twice the real density (49.5 notes per second: 0.33 of its 0.703), runs away in 31% of the rows and lets one instrument take over 16%; its density climbs inside the first window, before any restart (rows over 40 notes per second: 16% of its first 200 new notes, 46% of notes 447–894; real 13% and 18%). At T 0.6 its density stays flat except in 2 files that stack notes on one onset (most of its 85.9 notes per second), and it copies more (46%). Verbatim copies: musicar-L 14%, SkyTNT 14%, musicar-L-caption 0 (DRY).

Piano: 32 prompts, 1,000 new notes each



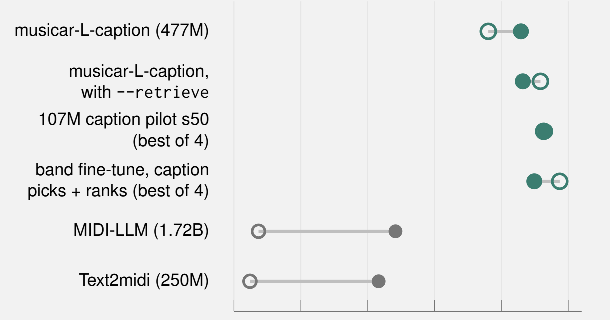
● the first 1,000 new notes, ○ the first 200. Solo-piano prompts from our held-out split of Aria-MIDI (Aria may have seen these files in training). For solo piano D reduces to notes per second and runaway.

The row metrics behind D (mean over the 32 rows)

system	notes / s	runaway	ended	copies	prompt check	D	D, 200	params
real continuation	7.95	0	0	0.001	1	0	0	
108M piano covers (ua_2048_v2, not released)	8.39	0	0.34	0.001	1	**0.018**	**0.006**	108M
musicar-L	8.95	0	0.41	0.003	1	**0.040**	**0.019**	464M
Aria medium, continuation [20]	9.99	0.031	0.31	0.007	1	0.096	0.067	659M
450M piano-cover fine-tune (not released)	3.53	0	1.00	0	1	0.177	0.078	464M

Aria runs with its authors' CUDA sampler at the README's settings (T 0.98, min-p 0.035), ours at T 0.9. Both continuation models of ours are closer to the real note rate than Aria: it plays faster (10.0 vs 8.0 notes per second) and runs away in 1 row of 32. The 450M piano-cover fine-tune is the one model of ours behind Aria: trained to write covers that end, it plays 3.5 notes per second and ends every row inside the 1,000 notes.

Text to MIDI: 27 captions + no caption, 16 pieces each, from the first note



● English pairs, ○ Ukrainian pairs. Every caption path of ours follows the captions better than both other open text-to-MIDI models, in both languages.

The scores behind F (mean over the 448 pieces of each system; 447 for MIDI-LLM and the band fine-tune: one file did not decode)

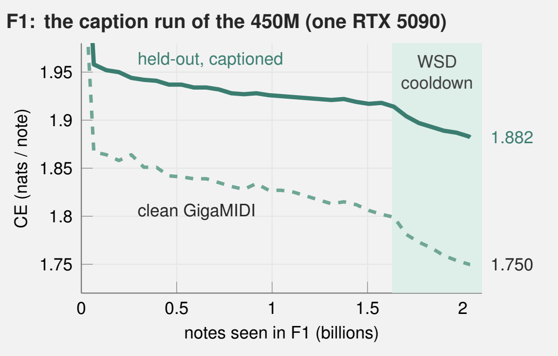
system	F, EN	F, UK	caption match	runaway	copies	notes / s	instr.	seconds	params
musicar-L-caption (app preset, 60 s)	1.07	0.95	0.857	0.021	0.031	8.8	9.5	63.5	477M
musicar-L-caption with `--retrieve` (60 s)	**1.08**	**1.15**	**0.871**	0.024	0.005	12.1	4.7	49.2	477M
107M caption pilot s50 (best of 4)	1.16	1.16	0.972	0.009	0.081	12.2	3.5	88.5	119M
band fine-tune, no text input (best of 4)	1.12	1.22	0.981	0.011	0.068	15.1	3.4	61.0	464M
MIDI-LLM [18]	0.60	0.09	0.553	0.141	0.021	24.6	3.0	35.0	1.72B
Text2midi [19]	0.54	0.06	0.554	0.090	0.016	23.3	11.4	25.2	250M

F over 9 English and 8 Ukrainian contrasts: "fast" / "slow", "minor" / "major", "loud" / "soft", "high" / "low" as caption pairs; "solo piano", "a drum beat", "without drums", "string ensemble", "electric guitar" each against no caption (no guitar prompt in Ukrainian). Caption match: see the glossary. Each model at its authors' sampling (MIDI-LLM T 1.0, top-p 0.98; Text2midi T 1.0); musicar-L-caption with the app's f1 preset and a 60-s length; with `--retrieve` the caption also picks a real 128-note GigaMIDI opening that it continues (not in the file, hence 49 s; for the empty prompt a random opening of the same files). s50 and the band fine-tune of musicar-L (multi_control: no text input, the caption picks the opening) keep the best of 4 candidates by caption match, so their rows are not single-sample scores. The 471M LaBSE encoder (frozen) is not counted in our parameters, Text2midi's FLAN-T5 encoder is.

Ukrainian. MIDI-LLM and Text2midi hardly move for Ukrainian captions (F 0.09 and 0.06): both learned text to MIDI from MidiCaps' English captions (MIDI-LLM: one Llama 3.2 1B with MIDI tokens added; Text2midi: FLAN-T5 + a decoder). musicar-L-caption reads both languages through LaBSE (F 0.95). They also run away more often (14% and 9% of 10-s windows over 40 notes per second vs 2%) and their pieces are shorter (35 and 25 s: no length input, they stop at their authors' token budgets; musicar-L-caption, asked for 60 s: 64 s).

② F1 and the caption phase: captions, sections and length on the 450M (2026-10-27 / 28)

F1: the caption run of the 450M (one RTX 5090)



Every 1,000 steps (65,536 notes per step). **Solid**: held-out captioned GigaMIDI; **dashed**: the base's clean validation set. Off the scale at step 0: 2.181 / 2.109 (the start, f1_plain512_long, was trained at 512 notes). Window conditions off in these evals.

The caption phase: F1 → f1_cap (600 steps at 512 notes)

	F1	f1_cap
held-out captioned CE (2,048 notes)	1.882	1.898
clean GigaMIDI CE	1.750	1.763
caption gain, notes 1–4 (vs none)	+0.377	+0.348
caption gain, notes 1–32 (vs none)	+0.072	+0.068
section gain (vs shuffled captions)	+0.0038	+0.0058
text follow F, guidance 1	0.77	0.82
text follow F, guidance 2	0.82	**1.03**
EN / UK pairs at guidance 2	0.79 / 0.86	1.07 / 0.97

In the pilots, the full recipe at 2,048 notes wore the caption reading down and a 512-note phase without spliced windows (≈60 captioned openings per step vs 13.8) rebuilt it. On F1 the reading was intact and the phase leaves it flat. Cost: +0.016 CE at 2,048 notes. Gain: caption following at guidance 2 (+0.20 F) and the 120-s length. f1_cap is musicar-L-caption.

Length control: 32 rows per length

asked for	F1	f1_cap	f1_cap, ramp 1*
nothing: median end	142 s	66 s	68 s
30 s: end / asked, within 20%	1.09, 72%	1.04, 75%	1.07, 72%
60 s	1.01, 84%	0.96, 81%	1.00, 75%
120 s	0.91, 44%	0.97, 56%	0.98, **81%**

End / asked = median over the rows that end themselves; within 20% = share of all rows. F1 and f1_cap columns: T 0.85, no top-p, no ramp; * EOS ramp 1, T 1.1, top-p 0.9. Without a length the caption phase writes shorter pieces (66 vs 142 s). With the EOS ramp (+1 logit per second over) and the search's sampling, 120-s pieces land best (81%); the ramp 0 / 0.5 runs at that sampling did not finish, so the ramp's own share is not measured. The app uses ramp 0.5 with a floor at 0.85 L and a stop at 1.2 L (not measured as one set).

Decoding settings: a search on f1_cap (2026-10-29)

24 settings × 8 rows, then the best 8 × 16 rows, then 3 × 32 rows of 2,048 notes; score J = F −0.25 Q, settings with runaway > 20% or copies > 25% out. **Winner** (the app's f1 preset): guidance 2.5, temperature 1.1, DRY on pitch-class chords (3, from 16 chords): F 1.07, F over the last notes 0.97, J 0.786. The defaults at 512 notes: J 0.34 and out (copies 0.48). Tested and not used: best of 4 / 8 (F 1.01 / 1.00 but D 2.13), a real opening picked by the caption (F 1.27 but copies 0.41), guidance against a neutral or the paired caption, guidance schedules, min-p. The F1 weights before the caption phase: J 0.30 with the best stage-A setting (f1_cap 0.62).

Holding the style of the opening (generation only)

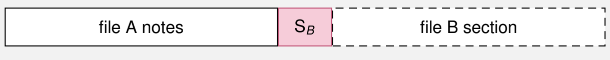
Long pieces drifted: the density and the instruments changed every section even with one caption. Keep-style turns the first 256 notes' own description into the section caption of the rest; lock-instruments keeps their instruments (+ families the caption names that have not played yet). 3 captions × 2 seeds × 120 s: busiest / calmest quarter 8.1× → 4.3× in notes per second, loudness swing 16 → 7 levels, instruments in the last quarter 10 → 4. "busy" / "dense" are not held above 25 notes per second: held, they pushed 1 piece in 6 into a runaway.

③ Section captions: what makes a later caption win (107M pilots)

Without these measures the model follows the first section's caption and then ignores later ones.

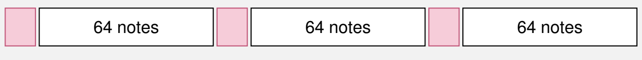
1. **Change-aware captions.** Each section caption names what changed since the section before ("drums enter, louder").

2. **Spliced windows (50% of windows).**



File A up to a section boundary, then a section of file B (dashed) with B's caption and no file caption: the caption contradicts the context and has to win.

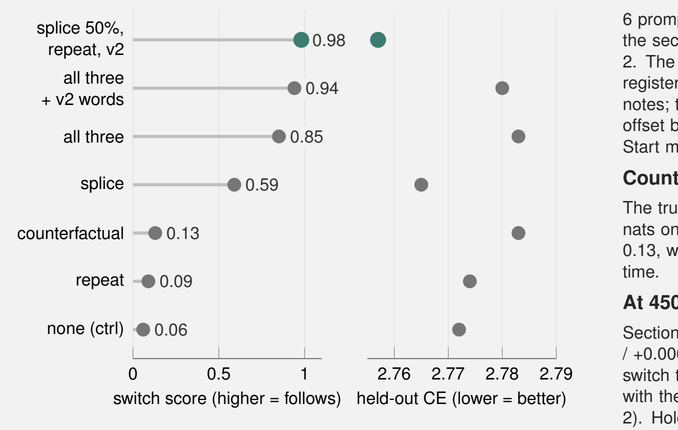
3. **Caption repeated every 64 notes,** in training and in generation.



4. **Section words from section statistics.** Loudness, pace and register words use thresholds over sections, not whole files ("loud" = the loudest 20% of sections), and pace is split into rhythm (onsets per second) and texture (notes per onset), so a section caption does not push the music to extremes.

5. **Caption dropout → guidance.** File captions are replaced by null vectors 15% of the time, section captions 20%, so generation can use classifier-free guidance [9].

Section-caption measures (107M pilot, 1,500–2,700 steps)



Each row continues the same model (v2_A0) for 1,500 steps with one measure or several; the last row runs 2,700 steps, the wall time of "all three" (no second forward). v2 words: measure 4. One seed, no error bars. **Teal** = the recipe in use.

Switch test

6 prompt pairs A/B (e.g. soft / loud), 16 samples each; the section caption changes at note 1,024, guidance 2. The matching attribute (velocity, notes per second, register, drum or string share) is measured per 256 notes; the score is its change after the switch over its offset before it (0 = keeps the first style, 1 = a full flip). Start model: 0.01.

Counterfactual loss [8]: not used

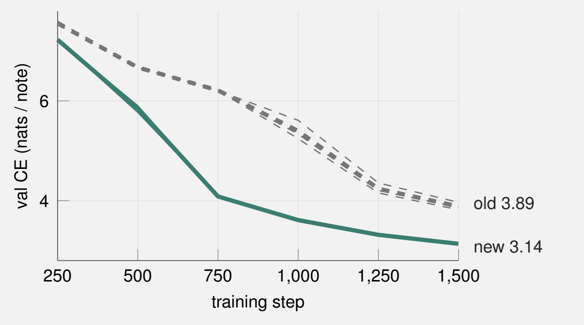
The true caption must beat the previous one by 0.05 nats on the changed section's first 32 notes: alone 0.13, with splice and repeat 0.85, at 1.8× the step time.

At 450M (F1)

Section captions beat shuffled ones by +0.004 (F1) / +0.006 (f1_cap) nats per note on held-out files; no switch test on F1 / f1_cap yet (the 450M rehearsal with the GQA merge showed no switch at guidance 2). Holding a caption still overshoots at 107M: asked for strings, 34% of samples run away.

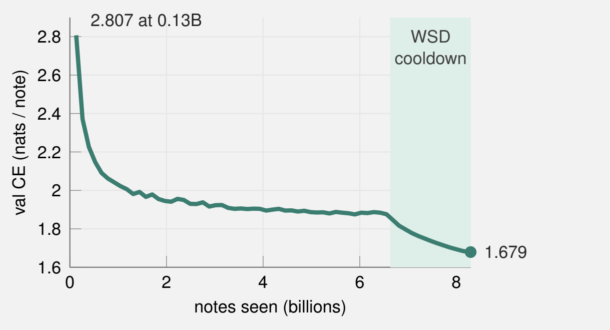
④ Why these choices: training curves (teal = what the model uses)

Trunk block (42M, 98M notes, 3 seeds each)



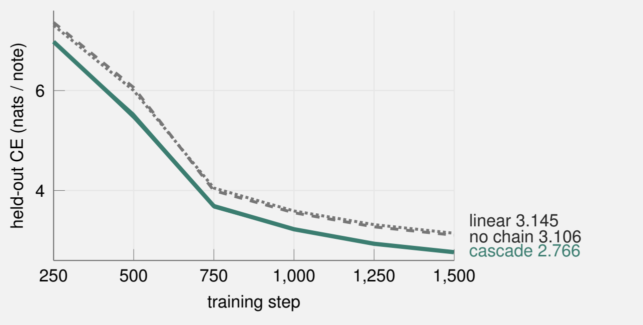
Clean GigaMIDI validation. **New**: RMSNorm, SwiGLU, QK-norm; **old** (dashed): LayerNorm, GELU; both RoPE and Muon. Thin lines = seeds, thick = mean; −0.75 at the end.

Base model pretraining (464M, 8.3B notes): musicar-L



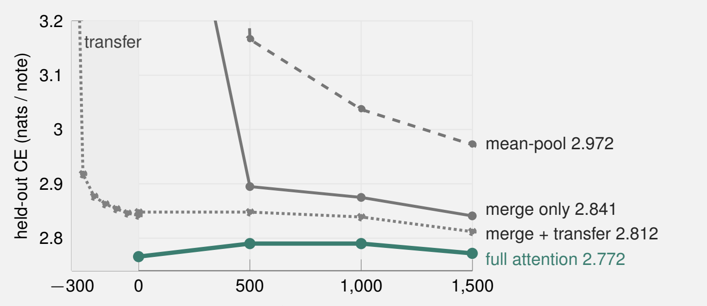
Clean GigaMIDI validation every 2,000 steps (65,536 notes per step); constant LR, then the $1-\sqrt{\cdot}$ cooldown in the last 20%. One run on one RTX 5090, ≈41 h.

Output heads (42M, 98M notes)



Held-out GigaMIDI. **Cascade**: each head reads the values already chosen, 3 seeds (thin) and mean. **No chain** (dashed): the same residual heads, each reading h_t alone; **linear** (dotted): one linear layer per attribute; one seed each.

GQA conversion (107M pilot, 12 → 4 key / value heads)



v2_A0 converted, then 1,500 steps on the section recipe's data, LR 10^{-4}, no section measures; the **teal** line is the same 1,500 steps without the conversion (full attention, what F1 uses). Off the scale at the start: mean-pool 10.15, merge 3.86–3.88. One seed.

GQA [10]: how a trained model is converted, and why F1 does not use it

What it buys. Each group of 3 query heads shares one key / value head: the K / V projections and the KV cache shrink 3×, so the 450M has 49.5M fewer parameters (464M → 415M without the caption adapters). Generation reads less per note: the 107M decodes 1.03× faster at 1 row and 2.28× at 64 rows (laptop GPU, bf16); on the Raspberry Pi (int4 weights, int8 cache) the estimate from bytes read per note is ≈1.3× at 1 row and ≈2.1× for 4 candidates with guidance.

Merge. Mean-pooling a group's K / V weights starts at CE 10.15. Instead, each group's value heads are merged by a PCA of their outputs weighted by the output projection, and the key heads per RoPE rotation pair by a rank-1 complex merge weighted by the query energy, so RoPE stays exact. Measured on 16 batches of the model's own inputs.

Transfer. 300 steps in which only the attention layers train, each to reproduce its own full-attention output (relative MSE) on the original layer inputs, LR 10^{-3}; the rest of the model is frozen.

Result at 107M. After 1,500 steps: mean-pool +0.200 vs full attention, merge only +0.069, merge + transfer +0.040 and still closing.

At 450M it failed. The 450M has QK-norm (the 107M pilots do not), and the merge flattened its attention: held-out CE 2.01 → 6.57 at the merge. An F1 rehearsal with the merge ended at 2.115 after 2,400 steps; the same recipe with full attention reached 2.058 in 600. GQA was not what kept the 450M from reading captions either. So F1 kept full attention; a conversion for the Raspberry Pi, if needed, is its own step (attention transfer first, CE within 0.05 of the captioned model before any further training).

Per head at step 1,500 (seed 1337, nats / note)

heads	Δt	pitch	dur.	vel.	sum	instr.
cascade	0.285	0.779	0.966	0.737	**2.766**	0.315
no chain	0.281	0.956	1.025	0.844	3.106	0.346
linear	0.286	0.970	1.037	0.853	3.145	0.353

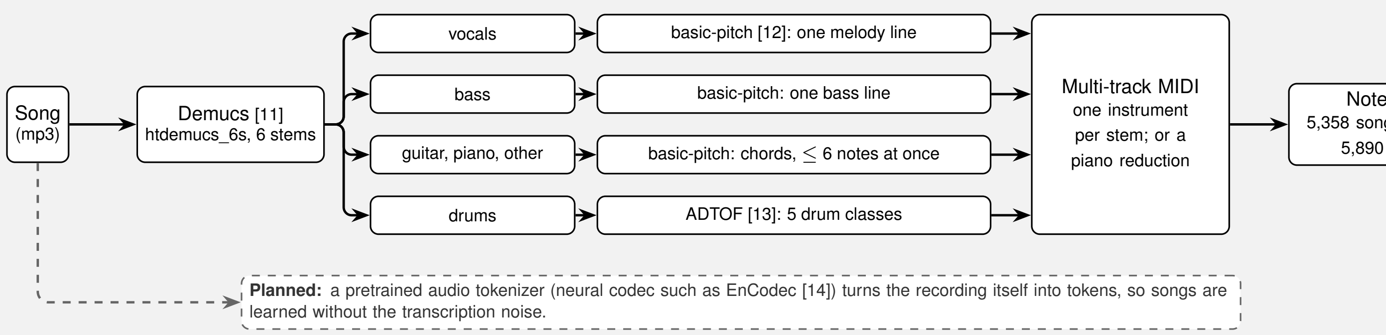
What the chain buys

The chain is worth −0.34 nats per note at the end (pitch +0.18, velocity +0.11 without it), the residual head form only −0.04. Δt is the first head, so it reads nothing extra and is the same in all three. Pitch gains most from the chain (−0.18): knowing when the note starts and on which instrument narrows its pitch. The sum is the held-out CE above; the instrument head is reported apart. An earlier 6M piano A/B (4 heads, chained plain MLP heads) had found only −0.03 for the chain: there its pitch head had lost the direct path from h_t.

⑤ Training stages

Setting	Base pretraining (done: musicar-L)	Caption training: F1 + caption phase (done: musicar-L-caption)	Fine-tune on Ukrainian songs
Start	from scratch	f1_plain512_long: the base + 1,500 steps of file captions at 512 notes, half the windows at a piece start; the caption phase starts from F1's last weights (GQA was tested and dropped: section 4)	the base (done, no captions); the captioned model (F2, planned)
Data	GigaMIDI [1] 0.4, Aria-MIDI [2] 0.3, Discover MIDI [3] 0.3; 8.3B notes (1.8 epochs on average; Aria ≈4.5)	captioned MIDI in 3 tiers: GigaMIDI with MidiCaps [4] / XMIDI [5] / GigaMIDI metadata / artist captions 0.55, Aria-MIDI with genre / composer 0.15, GigaMIDI unlabeled with feature captions 0.30; 2.04B notes	5,358 songs transcribed from audio (section 6; 4,811 train, 547 val) 0.75 + GigaMIDI replay 0.25 against forgetting
Captions	none: only the instrument-set and density conditions	templates filled from metadata (genre, mood, artist, composer) and from features computed from the notes (tempo, loudness, register, instruments, length), 30% in Ukrainian; MidiCaps' LLM-written captions; section captions from each section's own notes; windows with a caption get no instrument-set / density condition	none in the done run; F2 adds them
Window	2,048 notes (≈1–4 min)	2,048 notes + 16 file vectors + 4 per section caption arrival: 2,212 positions; half the windows at a piece start, a quarter at its end (EOS, length); caption phase: 512 notes, no spliced windows	2,048 notes
Batch	65,536 notes per step (32 windows)	65,536 notes per step: 32 windows of 2,048 in the long run, 128 of 512 in the caption phase	24,576 notes per step (12 windows)
Optimizer, LR	Muon [6] for weight matrices, AdamW for the rest; one LR 10^{-3} (Muon updates scaled to AdamW's size)	8-bit AdamW, LR 10^{-4}; caption adapters and length tables 10×	8-bit AdamW, LR $2 \cdot 10^{-4}$
Schedule	WSD [7]: warmup, stable (constant LR), $1-\sqrt{\cdot}$ decay over the last 20%	same; the caption phase: 100 warmup steps, decay over its last 30%	same
Steps	126,582 (≈41 h on one RTX 5090)	31,098 (≈14 h on one rented RTX 5090) + 600 at 512 notes (≈20 min)	2,000 (multi-track: the step-1,999 weights, multi_control)
Precision	bf16 with fp8 matmuls, torch.compile	same	same

⑥ Audio → MIDI: how songs become training notes (fine-tune data)


Song (mp3)
Demucs [11]
htdemucs_6s, 6 stems
vocals
bass
guitar, piano, other
drums
basic-pitch [12]: one melody line
basic-pitch: one bass line
basic-pitch: chords, ≤ 6 notes at once
ADTOF [13]: 5 drum classes
Multi-track MIDI
one instrument per stem; or a piano reduction
Note tokens
5,358 songs multi-track, 5,890 as piano

Planned: a pretrained audio tokenizer (neural codec such as EnCodec [14]) turns the recording itself into tokens, so songs are learned without the transcription noise.

Every step adds errors: stem bleed, missed or extra notes, octave and timing errors, drum classes mixed up, loudness guessed from activations; stems quieter than −30 dB vs the mix are dropped. The model learns these errors as if they were music.

⑦ Glossary

CE	cross-entropy of the true notes (nats per note), the sum over the note's heads	**D**	distance to real music: weighted sum over takeover, top share, notes/s (0.20 each), out of band, instruments (0.15), runaway (0.10) of |system − real| / the real rows' spread
held-out captioned	3,648 captioned GigaMIDI files kept out of training	**runaway**	continuation: share of rows above 40 notes per second; text and Q: share of 10-s windows above it
clean GigaMIDI	the base run's validation set (no captions)	**takeover**	one instrument plays over 80% of a row's notes
caption gain	CE without the caption (or with a shuffled one) minus CE with it, on the same notes	**out of band**	share of new notes on instruments the prompt did not use
switch score	how far a section caption moves the music after a switch (section 3)	**copies**	share of new pitched notes inside verbatim copies of $\geq$ 16 chords
F	text follow score: mean over caption pairs ("fast" vs "slow", ...) of the move in the asked direction over a clearly audible one (15 notes/s, 12 semitones, 8 velocity levels, 0.4 instrument share), clipped to [−1, 1.5]	**caption match**	0–1: mean over the categories a caption asks about (instruments, key, pace / loudness / register levels) of how well the piece meets them, half credit for near misses; 1 if nothing is checkable; judged with our caption builder's thresholds, so it favours our models
WSD [7]	warmup, stable, decay: LR schedule with a short final cooldown	**J**	F −0.25 Q; Q = D of the caption-free rows (no out-of-band term) + 4 × (runaway share over 0.10) + 4 × (copies over 0.25)
Muon [6]	optimizer that orthogonalises the update of each weight matrix	**GQA [10]**	grouped-query attention: groups of query heads share one key / value head
guidance [9]	classifier-free guidance: logits pushed away from the caption-free ones	**stem**	one separated part of a recording (vocals, bass, drums, ...)

⑧ References (links)

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